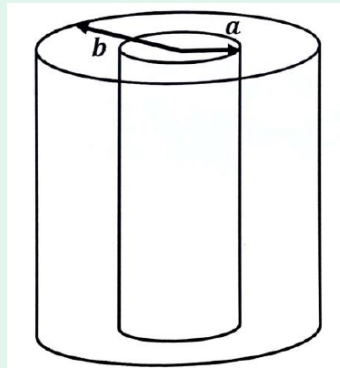


Problem 1 (SEE-the-LINE-DER): Consider the infinite cylindrical distribution below, given with the following charge density:

$$\rho(r) = \begin{cases} \rho, & r \leq a \\ \frac{\alpha}{r^2}, & a < r \leq b \end{cases}$$

With ρ as a uniform charge density. Calculate the electric field magnitude at the following regions:

- (a) $r < a$
- (b) $r = a$
- (c) $a < r < b$
- (d) $r = b$
- (e) $r > b$



Solution: The magnitude of an electric field $\vec{E}$ can be derived using Gauss's Law,

$$\Phi_E = \oint_{\text{GS}} \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E(2\pi rL) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E = \frac{Q_{\text{enc}}}{2\pi\epsilon_0 rL}$$

(a) At $r < a$, since the piecewise function states that there exists a uniform volume charge density,¹ then, this implies that there exists a charge within that region.

$$\begin{aligned} Q_{\text{enc}} &= \rho V_{\text{enc}} \\ &= \rho(\pi r^2 L) \end{aligned}$$

Substituting this into the formula above,

$$E = \frac{\rho\pi r^2 L}{2\pi\epsilon_0 rL} = \boxed{\frac{\rho r}{2\epsilon_0}}$$

¹Then we could let $\rho = Q/V$ since the charge is uniformly distributed throughout the surface.

(b) The magnitude of the electric field when $r = a$ is equal to the expression when $r < a$, but we let $r = a$,

$$E = \boxed{\frac{\rho a}{2\epsilon_0}}$$

(c) Let us first determine how to obtain the enclosed charge if $a < r < b$. Since it is bounded by the radii a and b , we need to consider the charge when $r' \leq a$ (inner region) and $a < r' < b$ (outer region) where r' is our observation point as $r' \rightarrow r$. Recall that,

$$\rho = \frac{dQ}{dV} \Rightarrow dQ = \rho dV$$

$$Q = \int dQ = \int \rho dV$$

In an infinite cylinder, we know that $dV = 2\pi r L dr$. Then,

$$\begin{aligned} Q_{\text{enc}} &= Q_{r' \leq a} + Q_{a < r' < b} \\ &= \int_0^a \rho(r') dV + \int_a^r \rho(r') dV \\ &= \int_0^a \rho(2\pi r' L) dr' + \int_a^r \frac{\alpha}{(r')^2} (2\pi r' L) dr' \\ &= \rho\pi L a^2 + 2\alpha\pi L \ln\left(\frac{r}{a}\right) \end{aligned}$$

Substituting this,

$$E = \boxed{\frac{\rho\pi L a^2 + 2\alpha\pi L \ln\left(\frac{r}{a}\right)}{2\pi\epsilon_0 r L}}$$

(d) When $r = b$, we simply let $r = b$ into the equation from (b), thus,

$$E = \boxed{\frac{\rho\pi L a^2 + 2\alpha\pi L \ln\left(\frac{b}{a}\right)}{2\pi\epsilon_0 b L}}$$

(e) This is simply the same as the equation we have solved from (c),

$$E = \boxed{\frac{\rho\pi L a^2 + 2\alpha\pi L \ln\left(\frac{r}{a}\right)}{2\pi\epsilon_0 r L}}$$

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Problem 2 (Face your Spheres): A solid conducting sphere with radius R carries a total charge $+Q$. The sphere is surrounded by an insulating shell with inner radius R and outer radius $2R$. The insulating shell has a non-uniform charge density $\rho(r) = \alpha r^2$, as given in the figure

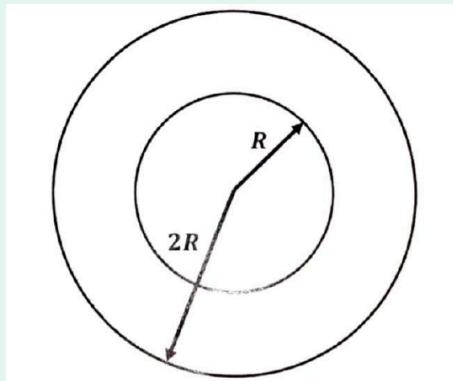
B. Calculate the electric field magnitude at the following regions:

- (a) $r < R$
- (b) $r = R$
- (c) $R < r < 2R$
- (d) $r = 2R$
- (e) $r > 2R$

C. If the volume charge density is instead given by

$$\rho(r) = \begin{cases} +Q, & r \leq R \\ \rho, & R < r \leq 2R \end{cases}$$

Where ρ is a uniform volume charge density, calculate the explicit expression of ρ so that the net charge of the entire system is zero.



Solution: The magnitude of an electric field $\vec{E}$ can be derived using Gauss's Law,

$$\Phi_E = \oint_{\text{GS}} \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E = \frac{Q_{\text{enc}}}{4\pi\epsilon_0 r^2}$$

(a) Recall that in a conductor the electric field inside it is 0, hence, when $r < R$,

$$E = \boxed{0}$$

(b) When $r = R$, we could imagine that we have two spheres that are drawn twice such that we could no longer distinguish if there are two spheres. Let us consider their charges as Q_{enc} which is simply the sum of the charge Q_{sphere} , contributed by the conductor and the charge Q_{shell} that is contributed by the insulator.

$$Q_{\text{enc}} = Q_{\text{sphere}} + Q_{\text{shell}}$$

Clearly $Q_{\text{sphere}} = +Q$. Let us figure out Q_{shell} using the given charge density $\rho(r)$,

$$\rho = \frac{dQ}{dV} \Rightarrow dQ = \rho dV$$

$$Q = \int dQ = \int \rho(r) dV$$

Recall that for a sphere, $dV = 4\pi r^2 dr$, let us denote r' as our observation point as $r' \rightarrow r$,

$$\begin{aligned} Q_{\text{shell}} &= \int_R^{r=R} \rho(r') dV = \int_R^{r=R} \alpha r'^2 (4\pi r'^2) dr' \\ &= \int_R^{r=R} 4\alpha\pi r'^4 dr' = 0 \end{aligned}$$

Thus, the conductor is the only one that contributes charge,

$$E = \boxed{\frac{Q}{4\pi\epsilon_0 R^2}}$$

(c) When $R < r < 2R$, we could use the same concept from (b). Since we have already created an expression for Q_{shell} , let us use that but we will solve it from R to r ,

$$\begin{aligned} Q_{\text{shell}} &= \int_R^r 4\alpha\pi r'^4 dr' = 4\pi\alpha \left(\frac{r^5}{5} - \frac{R^5}{5} \right) \\ Q_{\text{enc}} &= Q_{\text{sphere}} + Q_{\text{shell}} \\ \therefore Q_{\text{enc}} &= Q + 4\pi\alpha \left(\frac{r^5}{5} - \frac{R^5}{5} \right) \end{aligned}$$

Substituting this into the magnitude of an electric field equation simply gives us,

$$E = \frac{Q + 4\pi\alpha \left(\frac{r^5 - R^5}{5} \right)}{4\pi\epsilon_0 r^2} = \boxed{\frac{Q}{4\pi\epsilon_0 r^2} + \frac{\alpha(r^5 - R^5)}{5\epsilon_0 r^2}}$$

(d) Let $r = 2R$ in the equation from earlier,

$$\begin{aligned} Q_{\text{enc}} &= Q + 4\pi\alpha \frac{31R^5}{5} \\ E &= \boxed{\frac{Q + \frac{124\pi\alpha R^5}{5}}{16\pi\epsilon_0 R^2}} \end{aligned}$$

(e) We just need to modify the equation from (d),

$$E = \boxed{\frac{Q + \frac{124\pi\alpha R^5}{5}}{4\pi\epsilon_0 r^2}}$$

For the last problem, we need to find an expression of ρ such that the net charge $Q_{\text{net}} = 0$. This implies that,

$$Q_{\text{net}} = Q_{\text{sphere}} + Q_{\text{shell}} = 0$$

$$Q_{\text{shell}} = -Q_{\text{sphere}}$$

From the piecewise function, $Q_{\text{shell}} = +Q$, then we only need to figure out the charge of the shell Q_{shell} . When $R < r \leq 2R$, there exists a uniform volume charge density ρ , then,

$$\rho = \frac{Q_{\text{shell}}}{V_{\text{shell}}} \Rightarrow Q_{\text{shell}} = \rho V_{\text{shell}}$$

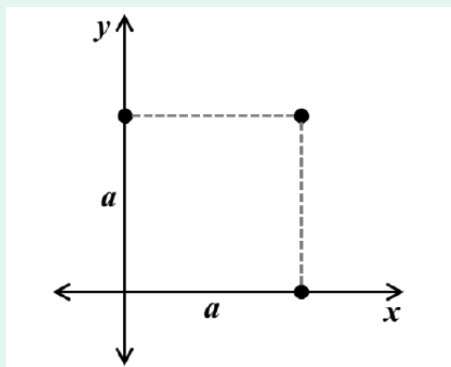
$$Q_{\text{shell}} = \rho \frac{4\pi}{3} [(2R)^3 - R^3] = \frac{\rho 28\pi R^3}{3}$$

$$= -(Q_{\text{sphere}}) = -Q$$

$$\therefore \rho = \boxed{-\frac{3Q}{28\pi R^3}}$$

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Problem 3 (Nah, I'd Work): Consider three electrons (each with charge $-|e|$) placed on the three corners of a square as shown. A proton ($+|e|$) is then brought from infinity to the origin.



- How much work must be done by an external force to bring the proton from infinity to the origin?
- What is the potential at the center of the square formed by these four charges?

Solution (a): Recall that the formula for bringing a charge from infinity,

$$\begin{aligned} W_{\text{ext}} &= -W_{a \rightarrow b} = U_b - U_a \\ &= U_b = kq_0 \sum_{i=0}^n \frac{q_i}{r_i} \end{aligned}$$

Let r_3 be this distance from the upper right corner to the origin,

$$r_3 = \sqrt{a^2 + a^2} = \sqrt{2}a$$

$$\begin{aligned}
 W_{\text{ext}} &= k|e| \left(\frac{-|e|}{a} + \frac{-|e|}{a} + \frac{-|e|}{\sqrt{2}a} \right) \\
 &= \frac{|e|}{4\pi\epsilon_0} \cdot \frac{-|e|(\sqrt{2} + \sqrt{2} + 1)}{\sqrt{2}a} \\
 &= \boxed{\frac{-|e|^2(2\sqrt{2} + 1)}{4\sqrt{2}\pi a\epsilon_0}}
 \end{aligned}$$

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Solution (b): Recall the formula for potential due to a collection of point charges,

$$\begin{aligned}
 V &= k \sum_{i=1}^n \frac{q_i}{r_i} = k \left(3 \cdot \frac{-|e|}{\sqrt{2}a} + \frac{|e|}{\sqrt{2}a} \right) \\
 &= \frac{1}{4\pi\epsilon_0} \left(\frac{-2|e|}{\sqrt{2}a} \right) = \boxed{-\frac{|e|}{2\sqrt{2}a\pi\epsilon_0}}
 \end{aligned}$$

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